Machine Learning Classification of High-Protein Foods Using Nutrient Data

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Course Code: DATA7703

Course Name: Machine Learning

Submission Date: 28 May 2026

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1.Introduction

Machine learning techniques are widely used to identify patterns and relationships within large datasets. In food and nutrition research, nutrient composition data can contain important information that may help classify foods into different nutritional categories. This assignment investigates whether nutrient values can be used to classify foods as high in protein without directly using the protein value itself as a predictor.

The dataset used in this investigation was the Australian Food Composition Database (AFCD) Release 3 nutrient dataset. The dataset contains nutrient composition information for a wide range of foods measured per 100 grams. Since the dataset includes many nutritional variables such as fats, carbohydrates, vitamins, minerals and amino acids, it provides a suitable dataset for applying machine learning classification techniques.

The main objective of this assignment was to determine whether other nutrient variables could successfully predict whether a food is high in protein. To achieve this, several machine learning models were developed and compared, including Logistic Regression, k-Nearest Neighbours (k-NN), and Random Forest. The performance of these models was evaluated using metrics such as accuracy, precision, recall and F1-score.

The assignment also involved preprocessing the dataset, handling missing values, exploratory data analysis, feature importance analysis and hyperparameter tuning. The overall aim was not only to achieve strong predictive performance, but also to understand which nutrient variables contributed most strongly to identifying high-protein foods.

2.Research Question and Objectives

The main research question investigated in this assignment was:

Can nutrient values be used to classify whether a food is high in protein without directly using the protein value itself?

To answer this question, a binary classification problem was created using the nutrient dataset. Foods with a protein value greater than or equal to 10 g per 100 g were classified as “high protein”, while foods below this threshold were classified as “not high protein”. This threshold was selected as a reasonable assumption for separating foods with relatively high protein content from foods with lower protein content.

Several objectives were established for this investigation:

1. Preprocess and clean the nutrient dataset for machine learning analysis.
2. Explore relationships and patterns within the nutrient variables using exploratory data analysis (EDA).
3. Develop and compare multiple machine learning classification models.
4. Evaluate the performance of the models using classification metrics.
5. Identify which nutrient variables contribute most strongly to predicting high-protein foods.
6. Investigate whether hyperparameter tuning improves model performance.

The machine learning models selected for comparison were Logistic Regression, k-Nearest Neighbours (k-NN), and Random Forest. These models were chosen because they represent different machine learning approaches, including linear models, distance-based learning and ensemble learning methods.

3.Dataset Description

The dataset used for this assignment was the Australian Food Composition Database (AFCD) Release 3 nutrient dataset. This dataset contains nutritional information for a large variety of foods commonly consumed in Australia. The nutrient values are reported per 100 grams of food, allowing foods to be compared consistently across the dataset.

For this investigation, only the “All solids & liquids per 100 g” worksheet was used to maintain consistency in the measurements across all food items. After loading the dataset into Python using the pandas library, the dataset initially contained 1588 rows and 272 columns.

Several columns were not useful for machine learning analysis and were removed during preprocessing. These included identifier and descriptive columns such as:

- Public Food Key
- Classification
- Derivation
- Food Name

The protein column was used only to create the target variable and was then removed from the predictors to prevent data leakage. The nitrogen variable was also removed because nitrogen is strongly related to protein content and could unfairly influence the models.

A binary target variable called High_Protein was created:

- Foods with protein values greater than or equal to 10 g per 100 g were labelled as 1 (high protein).
- Foods with protein values below 10 g per 100 g were labelled as 0 (not high protein).

After creating the target variable and removing unnecessary columns, the dataset was analysed for missing values. Many nutrient variables contained extremely high percentages of missing data, with some columns containing more than 99% missing values. To reduce noise and improve model reliability, columns with more than 70% missing values were removed from the dataset.

The remaining missing numeric values were handled using median imputation. Median imputation was selected because it is less sensitive to extreme values and skewed distributions compared to mean imputation. After preprocessing, the final cleaned dataset contained 1588 rows and 146 columns with no remaining missing values.

4.Data Preprocessing

Data preprocessing was an important stage of this assignment because the raw nutrient dataset contained missing values, unnecessary columns and variables that could potentially introduce data leakage into the machine learning models.

The dataset was first loaded into Python using the pandas library. During initial inspection, several unnamed columns were identified which appeared to come from formatting within the Excel spreadsheet. These columns did not contain useful information and were removed from the dataset.

The protein variable was used to create the binary target variable High_Protein. However, after the target variable was created, the original protein column was removed from the predictor variables. This was done to avoid data leakage, where the model would otherwise directly learn the answer from the protein values themselves rather than learning relationships between other nutrients and protein content. The nitrogen variable was also removed because nitrogen is strongly associated with protein and could similarly introduce leakage into the model.

After the unnecessary columns were removed, the dataset was analysed for missing values. The analysis showed that many nutrient variables had extremely large percentages of missing data, with several columns containing more than 90% missing values. These highly incomplete columns were removed from the dataset using a threshold of 70% missing values. This decision reduced noise and simplified the dataset while retaining the majority of useful nutrient variables.

The remaining missing numeric values were handled using median imputation. Median imputation was selected because nutrient data often contains skewed distributions and outliers. Compared to mean imputation, the median is less influenced by extreme values and therefore provides a more robust estimate for missing data.

After preprocessing, the final cleaned dataset contained 1588 food samples and 146 columns with no remaining missing values. The cleaned dataset was then used for exploratory data analysis and machine learning model development.

5.Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to better understand the structure of the dataset and identify relationships between nutrient variables and the target variable. Several visualisations were created to analyse class distributions, feature relationships and correlations between nutrients.

Target Variable Distribution

The distribution of the target variable High_Protein was first examined to determine whether the dataset was imbalanced. The analysis showed that approximately 57% of the foods were classified as not high protein, while around 43% were classified as high protein.

This indicates that the dataset is only mildly imbalanced, meaning that the machine learning models were unlikely to become heavily biased toward one class. Because the imbalance was relatively small, additional balancing techniques such as oversampling or SMOTE were not required.

Correlation Heatmap

A correlation heatmap was generated to visualise relationships between nutrient variables. The heatmap showed several groups of highly correlated features, particularly among fatty acids, vitamins and mineral-related variables.

The presence of highly correlated features suggests that some nutrients contain overlapping information. This is important because certain machine learning models, particularly linear models, can be affected by multicollinearity. However, ensemble models such as Random Forest are generally more robust to correlated predictors.

The correlation heatmap also demonstrated the complexity of the nutrient dataset and justified the use of machine learning models capable of handling large numbers of interacting variables.

Correlation with High Protein Classification

The correlation analysis showed that several nutrients had relatively strong positive relationships with the High_Protein target variable. Some of the strongest correlated features included:

- Tryptophan
- Phosphorus
- Zinc
- Selenium
- Cholesterol
- Niacin-related variables

These findings are nutritionally reasonable because many protein-rich foods such as meat, fish, eggs, dairy and legumes also contain high levels of amino acids, minerals and vitamins. This suggests that the machine learning models were identifying meaningful nutritional relationships rather than relying on random patterns in the data.

Overall, the exploratory analysis provided useful insight into the dataset and helped justify the later modelling decisions used throughout the assignment.

6.Logistic Regression Model

Introduction

Logistic Regression was selected as the first machine learning model for this investigation. Logistic Regression is a supervised classification algorithm commonly used for binary classification problems. The model predicts the probability that an observation belongs to a particular class using a linear combination of input features.

This model was chosen because it provides a relatively simple and interpretable baseline for comparison against more complex machine learning methods. Since the target variable in this assignment contains two classes (High_Protein = 1 and High_Protein = 0), Logistic Regression was suitable for the classification task.

Data Scaling

Before training the Logistic Regression model, the dataset was standardised using StandardScaler from the scikit-learn library. Standardisation transforms the features so they have approximately zero mean and unit variance.

Feature scaling was necessary because nutrient variables exist on very different scales. For example, some vitamins and minerals are measured in micrograms while other nutrients such as carbohydrates or fats are measured in grams. Without scaling, variables with larger numeric ranges could dominate the model.

The dataset was split into training and testing sets using an 80/20 split. Stratification was applied to preserve the class distribution of the target variable across both datasets.

Model Performance

The Logistic Regression model achieved strong classification performance on the unseen test dataset. The results were:

- Accuracy: 95.6%
- Precision: 94.8%
- Recall: 94.8%
- F1-score: 94.8%

The confusion matrix showed that the model produced relatively few incorrect predictions, with only small numbers of false positives and false negatives.

The strong performance suggests that there are meaningful linear relationships between nutrient variables and whether a food is classified as high in protein. Even after removing the protein variable itself, the model was still able to identify high-protein foods with high accuracy using other nutrient information.

One limitation of Logistic Regression is that it assumes relatively linear relationships between predictors and the target variable. Although the model performed very well, more complex models may be able to capture non-linear nutrient interactions more effectively.

7.k-Nearest Neighbours (k-NN) Model

Introduction

The second machine learning model used in this investigation was k-Nearest Neighbours (k-NN). Unlike Logistic Regression, k-NN is a distance-based machine learning algorithm that classifies observations based on the labels of neighbouring data points.

The model works by identifying the nearest neighbours to a new observation and assigning the majority class among those neighbours. In this assignment, the number of neighbours was set to 7 (k = 7).

k-NN was selected because it provides a useful comparison against Logistic Regression. While Logistic Regression attempts to learn a linear decision boundary, k-NN relies directly on distances between observations in the feature space.

Feature Scaling

Since k-NN depends heavily on distance calculations, feature scaling was essential before training the model. StandardScaler was again applied to standardise all nutrient variables.

Without scaling, nutrients measured on larger numeric scales could dominate the Euclidean distance calculations and negatively affect the model's performance.

Model Performance

The k-NN model achieved the following results on the unseen test dataset:

- Accuracy: 91.2%
- Precision: 93.5%

- Recall: 85.2%
- F1-score: 89.1%

Although the model performed reasonably well overall, its performance was noticeably lower than Logistic Regression. In particular, the recall score was lower, indicating that the model missed more high-protein foods during classification.

The confusion matrix showed a larger number of false negatives compared to Logistic Regression. This suggests that the model struggled more with correctly identifying all high-protein food items.

One possible explanation for the weaker performance is the high dimensionality of the dataset. Even after preprocessing, the dataset still contained 145 predictor variables. Distance-based algorithms such as k-NN can struggle in high-dimensional feature spaces because distances between observations become less meaningful. This problem is commonly referred to as the “curse of dimensionality”.

Another limitation of k-NN is that the algorithm is sensitive to noisy features and irrelevant variables. Since the nutrient dataset contains many correlated nutrient variables, the model may have been affected by overlapping information between predictors.

Overall, the k-NN model demonstrated acceptable performance, but it did not perform as strongly as Logistic Regression or Random Forest for this dataset.

8.Random Forest Model

Introduction

The final machine learning model used in this investigation was Random Forest. Random Forest is an ensemble learning method that combines multiple decision trees to improve predictive performance and reduce overfitting.

Unlike Logistic Regression, Random Forest is capable of modelling complex non-linear relationships between variables. It is also more robust to correlated features and noisy data, making it well suited for large nutritional datasets with many interacting nutrient variables.

Another advantage of Random Forest is that it provides feature importance scores, allowing interpretation of which nutrient variables contribute most strongly to the classification task.

Model Training

The Random Forest model was trained using the original unscaled dataset because tree-based models do not require feature standardisation. The initial model used 200 decision trees (n_estimators = 200) with a fixed random state for reproducibility.

The same 80/20 train-test split used in previous models was retained to ensure fair comparison across all machine learning methods.

Initial Model Performance

The initial Random Forest model achieved the following results on the test dataset:

- Accuracy: 96.5%
- Precision: 95.6%

- Recall: 96.3%
- F1-score: 95.9%

These results were slightly better than Logistic Regression and substantially better than the k-NN model. The confusion matrix showed that the Random Forest model produced very few incorrect predictions and classified both classes consistently well.

The stronger performance of Random Forest suggests that the nutrient dataset contains complex interactions and non-linear relationships between variables. Random Forest was able to capture these relationships more effectively than the simpler Logistic Regression model.

Feature Importance Analysis

One of the most useful aspects of Random Forest is its ability to estimate feature importance. The analysis showed that several nutrient variables contributed strongly to predicting whether a food was high in protein.

Some of the most important features included:

- Tryptophan
- Phosphorus
- Selenium
- Zinc
- Cholesterol
- Niacin-related nutrients

These results are nutritionally meaningful because foods high in protein are often rich in amino acids, minerals and vitamins. The feature importance analysis therefore suggests that the model was learning realistic nutritional relationships rather than relying on random correlations.

The feature importance graph also demonstrated that some nutrients contributed far more strongly than others, indicating that not all variables were equally useful for classification.

Overall, Random Forest produced the strongest performance among the initial machine learning models and provided the most interpretable insight into which nutrients influenced the predictions.

9.Hyperparameter Tuning

After evaluating the initial Random Forest model, hyperparameter tuning was performed to investigate whether model performance could be further improved. Hyperparameter tuning is an important step in machine learning because the choice of model parameters can significantly influence predictive performance.

GridSearchCV from the scikit-learn library was used to perform systematic hyperparameter tuning with 5-fold cross-validation. Cross-validation helps provide a more reliable estimate of model performance by repeatedly splitting the training data into different training and validation subsets.

The following hyperparameters were tested during the tuning process:

- Number of trees (n_estimators)

- Maximum tree depth (max_depth)
- Minimum samples required to split a node (min_samples_split)

Several combinations of parameter values were evaluated to identify the model configuration that produced the highest F1-score.

The best-performing parameter combination was:

- n_estimators = 200
- max_depth = None
- min_samples_split = 5

The tuned model achieved a cross-validation F1-score of approximately 0.960.

After retraining the model using the optimal parameters, the tuned Random Forest model achieved the following performance on the unseen test dataset:

- Accuracy: 97.2%
- Precision: 96.3%
- Recall: 97.0%
- F1-score: 96.7%

These results represent a small but meaningful improvement over the original Random Forest model. The tuning process slightly reduced classification errors and improved the balance between precision and recall.

The improvement demonstrates that hyperparameter tuning can help optimise machine learning models by adjusting how the model learns patterns from the data. It also highlights the importance of experimentation and validation when developing predictive models.

Overall, the tuned Random Forest model produced the strongest performance of all models tested in this assignment.

10.Final Model Comparison and Discussion

The performance of all machine learning models developed during this investigation was compared using accuracy, precision, recall and F1-score. The final results are summarised in Table 1.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	95.6%	94.8%	94.8%	94.8%
k-NN	91.2%	93.5%	85.2%	89.1%
Random Forest	96.5%	95.6%	96.3%	95.9%
Tuned Random Forest	97.2%	96.3%	97.0%	96.7%

The results show that all machine learning models were capable of classifying high-protein foods reasonably well. However, there were clear differences in model performance.

The k-NN model produced the weakest results overall, particularly in terms of recall. This indicates that the model missed a larger number of high-protein foods compared to the other models. One likely reason is the high dimensionality of the nutrient dataset. Distance-based algorithms such as k-NN can struggle when many predictor variables are present because distances between observations become less informative.

Logistic Regression performed substantially better than k-NN. This suggests that there are strong linear relationships between nutrient composition and whether a food is high in protein. The model achieved balanced precision and recall scores and produced relatively few classification errors.

The Random Forest models achieved the strongest overall performance. This indicates that the nutrient dataset likely contains more complex and non-linear relationships that tree-based ensemble methods can capture effectively. The tuned Random Forest model achieved the highest accuracy and F1-score after hyperparameter optimisation, demonstrating the value of model tuning and experimentation.

Another important outcome from this investigation was the feature importance analysis. Nutrients such as tryptophan, phosphorus, selenium and zinc were identified as highly important predictors. These nutrients are commonly associated with protein-rich foods such as meat, dairy products, seafood and legumes. This suggests that the machine learning models identified meaningful nutritional relationships within the dataset.

Overall, the results strongly support the original research question. Even after removing the protein variable itself, the models were still able to classify foods as high in protein with very high accuracy using only the remaining nutrient variables.

11.Reflection

This assignment helped improve my understanding of several important machine learning concepts, particularly data preprocessing, feature selection and model evaluation. Before starting this assignment, I underestimated how important preprocessing decisions are when developing machine learning models. During this investigation, it became clear that handling missing values, removing noisy variables and preventing data leakage can significantly affect model performance.

One of the most important things I learned was the importance of avoiding data leakage. Initially, it seemed reasonable to keep all nutrient variables in the dataset, but I realised that variables such as protein and nitrogen could allow the models to indirectly learn the answer rather than identify meaningful relationships between other nutrients and protein-rich foods. Removing these variables made the classification task more realistic and improved my understanding of proper machine learning practice.

Another concept I understood better through this assignment was the effect of different model types on structured datasets. I originally expected k-NN to perform similarly to the other models, but the results showed that it struggled more with the high-dimensional nutrient data. This helped me better understand the curse of dimensionality and why distance-based methods can become less effective when many features are present.

The Random Forest model was particularly interesting because it not only achieved the strongest performance, but also provided feature importance information. This made the model easier to interpret and showed how certain nutrients are strongly associated with high-protein foods. I found it interesting that nutrients such as tryptophan, phosphorus and zinc appeared as important predictors because these nutrients are commonly found in protein-rich foods.

Hyperparameter tuning was another useful learning experience. Before this assignment, I had limited understanding of how tuning affects machine learning performance. Using GridSearchCV and cross-validation helped me understand how different parameter combinations can slightly improve predictive accuracy and reduce errors.

One challenge encountered during the assignment was managing the large number of nutrient variables and missing values within the dataset. Choosing appropriate thresholds for removing missing data required experimentation and judgement rather than simply following fixed rules.

Overall, this assignment improved my confidence in applying machine learning workflows from preprocessing and exploratory analysis through to model training, evaluation and interpretation. It also helped connect practical coding work with concepts discussed during lectures and practical classes.

12.Conclusion

This assignment investigated whether nutrient values could be used to classify foods as high in protein without directly using the protein variable itself. Several machine learning models were developed and evaluated using the Australian Food Composition Database nutrient dataset.

The investigation showed that nutrient variables contain strong predictive information related to protein-rich foods. Even after removing the protein and nitrogen variables to prevent data leakage, the machine learning models were still able to achieve high classification performance.

Among the models tested, the tuned Random Forest model produced the best overall performance with an accuracy of approximately 97.2% and an F1-score of 96.7%. Logistic Regression also performed strongly, while the k-NN model produced weaker results due to the high-dimensional nature of the dataset.

The feature importance analysis identified nutrients such as tryptophan, phosphorus, selenium and zinc as important predictors of high-protein foods. These findings were nutritionally reasonable and demonstrated that the models learned meaningful relationships from the dataset.

Overall, the results successfully answered the research question and demonstrated the effectiveness of machine learning methods for nutritional classification tasks. The assignment also highlighted the importance of preprocessing, model selection and hyperparameter tuning when developing machine learning solutions.

Figure 1 — Target Distribution Plot

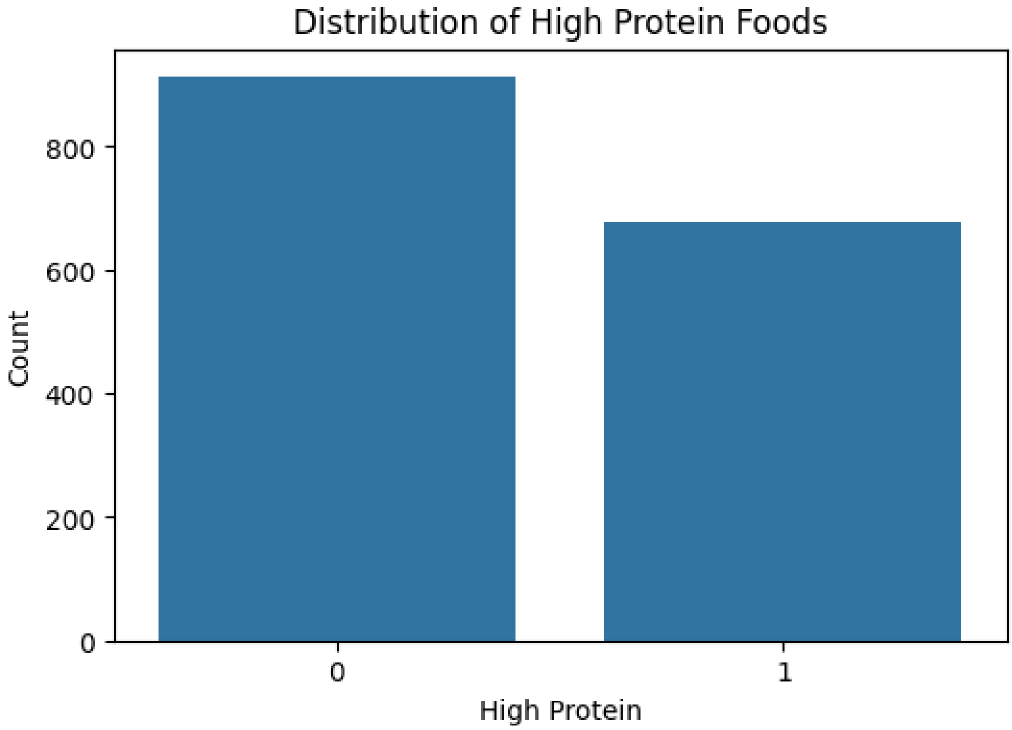

Figure 1: Distribution of high-protein and non-high-protein foods.

This figure shows the class distribution of the target variable High_Protein. The dataset is only mildly imbalanced, with slightly more foods classified as not high protein than high protein. This indicates that severe class imbalance handling techniques were not necessary for this investigation.

Figure 2 — Correlation Heatmap

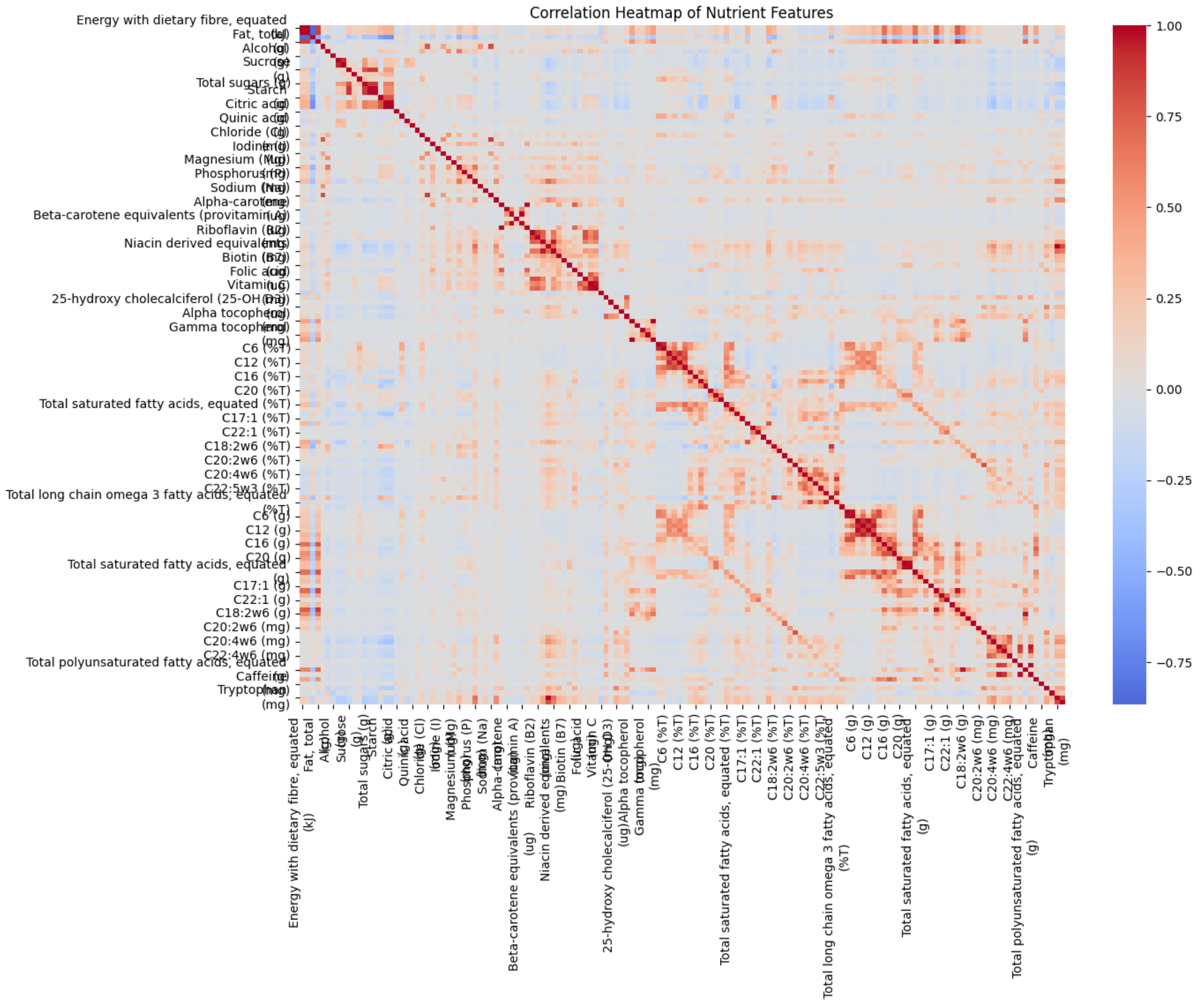

Figure 2: Correlation heatmap of nutrient variables after preprocessing.

This heatmap visualises relationships between nutrient variables within the cleaned dataset. Several groups of highly correlated nutrient features can be observed, particularly among vitamins, minerals and fatty acid variables. The presence of correlated variables suggests overlapping nutritional information within the dataset.

Figure 3 — Logistic Regression Confusion Matrix

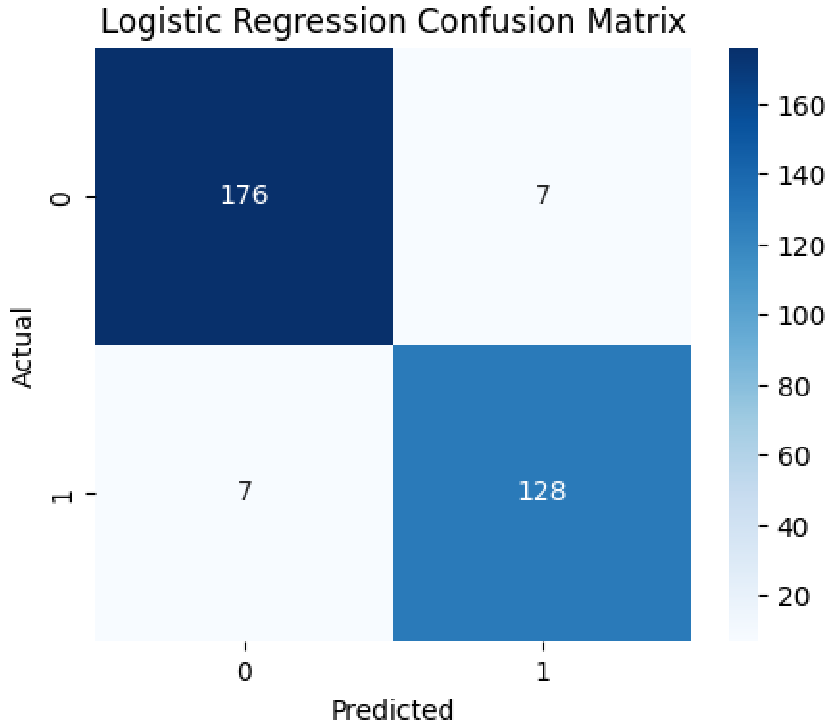

Figure 3: Logistic Regression confusion matrix.

The confusion matrix for Logistic Regression shows that the model achieved strong classification performance with relatively few incorrect predictions. The model demonstrated balanced performance across both high-protein and non-high-protein food classes.

Figure 4 — k-Nearest Neighbours (k-NN) Confusion Matrix

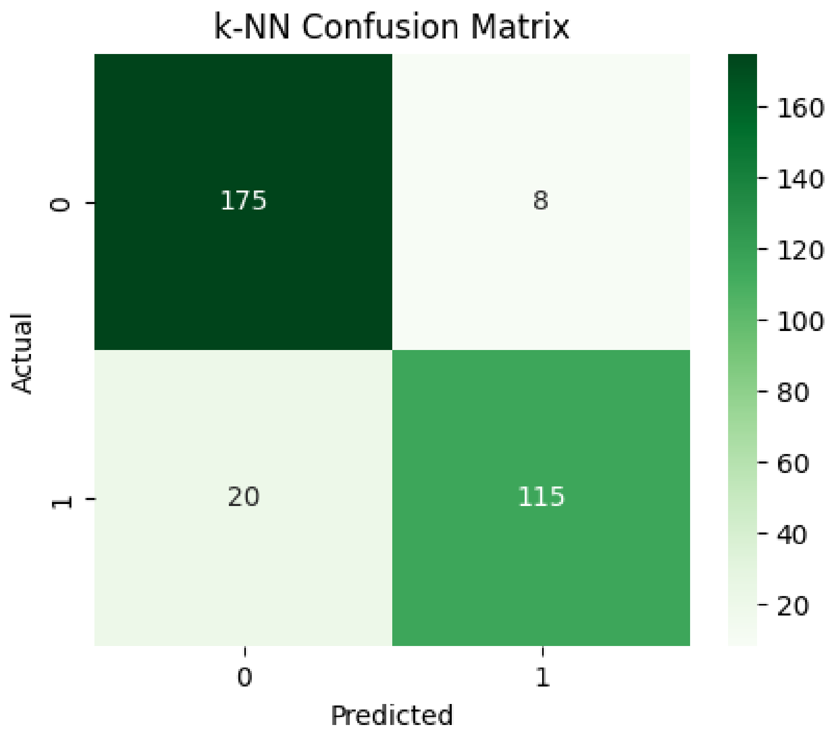

Figure 4: k-NN confusion matrix.

The k-NN confusion matrix shows that the model produced more classification errors compared to Logistic Regression and Random Forest. In particular, the model generated more false negatives, meaning some high-protein foods were incorrectly classified as not high protein.

Figure 5 — Random Forest Confusion Matrix

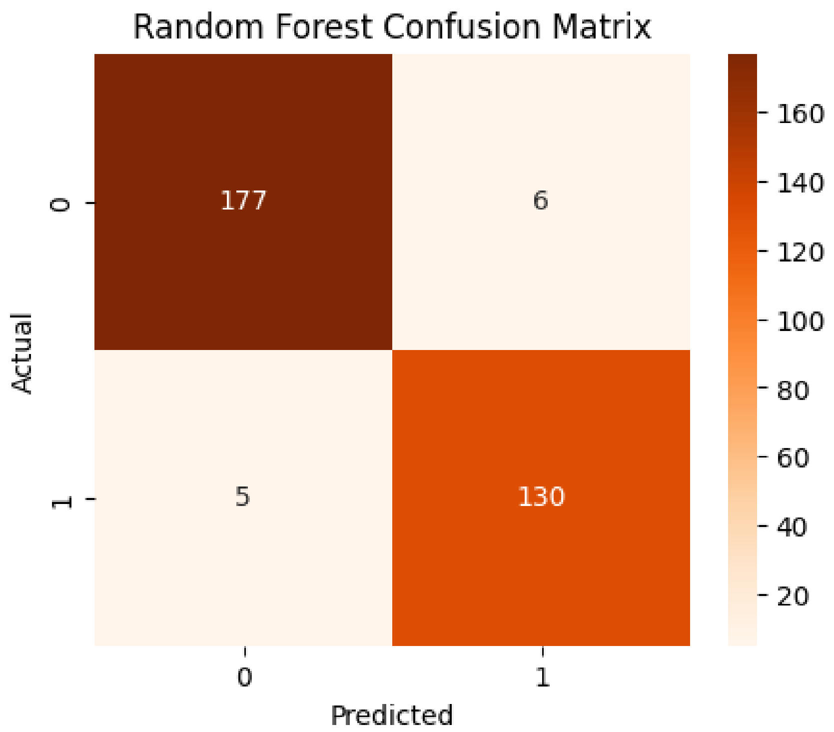

Figure 5: Random Forest confusion matrix.

The Random Forest model produced the strongest classification performance among the initial machine learning models. The confusion matrix shows very few incorrect predictions and strong classification consistency across both classes.

Figure 6 — Feature Importance Plot

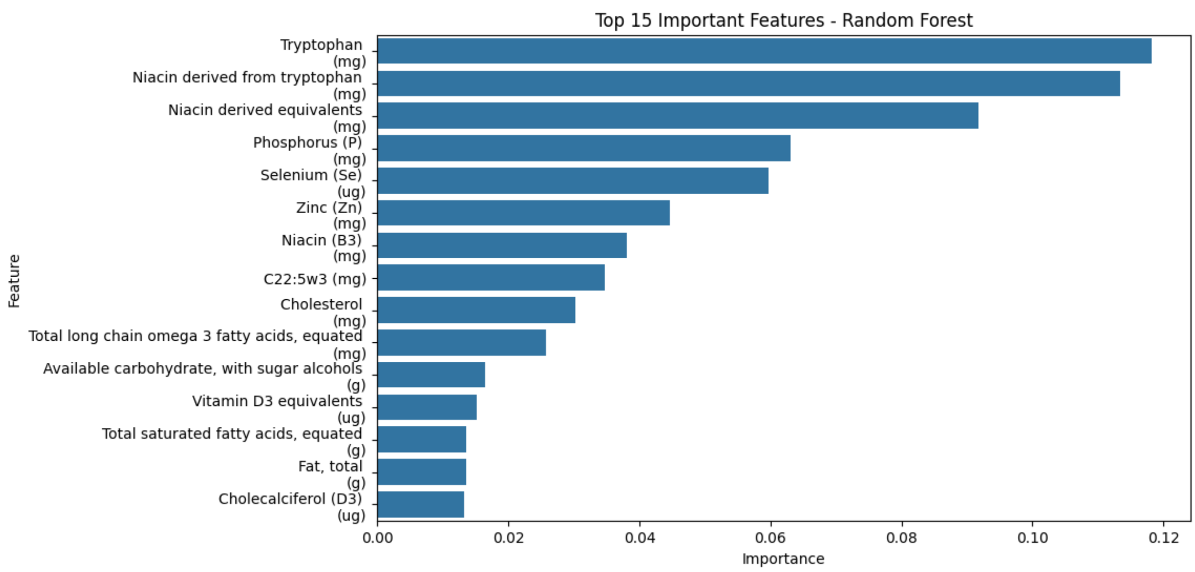

Figure 6: Top 15 feature importance scores from the Random Forest model.

This figure displays the most important nutrient variables identified by the Random Forest model. Nutrients such as tryptophan, phosphorus, selenium and zinc contributed strongly to predicting whether foods were classified as high in protein.